Aquaculture Reports 39 (2024) 102510

Contents lists available at ScienceDirect

Aquaculture Reports

journal homepage: www.elsevier.com/locate/aqrep

The effect of the *Hippocampus erectus* decoction on improving osteoporosis in zebrafish

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A R T I C L E I N F O

Keywords:
Osteoporosis
Zebrafish larvae
Seahorse decoction
Prednisolone

A B S T R A C T

The seahorse is a traditional Chinese herbal medicine that has been utilized for the treatment of conditions such as osteoporosis, nonunion fractures, knee osteoarthritis, and herniated lumbar disks.

Zebrafish is an excellent animal model to study the effects of herbal preparations on osteogenesis and bone metabolism, and bone biology is highly similar to that of mammals, so it has been used to develop multiple glucocorticoid-induced osteoporosis (GIOP) models. Our study aimed to explore whether the decoction of *Hippocampus erectus* (DHe) administered in zebrafish larvae promotes bone health under physiological and osteoporotic conditions. In zebrafish larvae, 25 μM prednisolone solution was successfully used to induce the zebrafish osteoporosis model, which can significantly inhibit the mineralization area and degree of bone mineralization, reduce the activity of alkaline phosphatase, and increase tartrate-resistant acid phosphatase activity. The results showed that 0.3 g/L DHe had a significant improvement effect on osteoporosis in zebrafish by enhancing the mineralization degree and phosphatase activity of zebrafish, while also promoting the expression of genes associated with bone development. It can be considered a good candidate to develop new drugs against osteoporosis.

1. Introduction

Osteoporosis is a metabolic bone disease, with defining features including decreased bone density, heightened bone fragility, damaged bone microstructure, and increased susceptibility to fractures (Anam and Insogna, 2021). The increasing pace of global aging will contribute to a further escalation in the prevalence of osteoporosis, accompanied by the gradual worsening of symptoms. Glucocorticoids are a leading cause of secondary osteoporosis, however, the current pharmacological treatments for human bone diseases have some side effects, including upper gastrointestinal symptoms and bowel disturbances (Tarantino et al., 2017). So it is important to find innovative non-invasive natural therapeutic approaches. Some traditional Chinese herbs have antioxidant and anti-inflammatory properties, which have been proven to play an important role in preventing skeletal diseases (Carnovali et al., 2021, 2020).

The seahorse is a valuable medicine that has been widely used for centuries in traditional Chinese medicine. Studies have shown that seahorses are rich in various chemical components such as amino acids, fatty acids, and trace elements (Lin et al., 2008; Sanaye et al., 2014), which have the properties of appetite enhancement, antioxidant, anti-tumor, anti-aging, and anti-fatigue (Jianying et al., 2000; Zhang et al., 2003), and have a curative role for osteoporosis, impotence, infertility, kidney disorders and skin afflictions (Kumaravel et al., 2012; Rosa et al., 2013; Zhang et al., 2003). The phenolic compounds, extracted from the seahorses, can promote the up-regulation of osteogenic marker genes, the proliferation and mineralization of osteoblasts, induce the formation of bone nodules, and inhibit the activity of tartrate-resistant acid phosphatase (TRAP) in osteoclasts in rats (Li et al., 2019; Yuan et al., 2018).

The novel peptides isolated from the seahorses can promote the differentiation of osteoblasts and chondrocytes, effectively restrain the

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https://doi.org/10.1016/j.aqrep.2024.102510
Received 5 July 2024; Received in revised form 26 September 2024; Accepted 12 November 2024
Available online 15 November 2024

increase of matrix metalloprotein, and refrain the breakdown of collagen by inhibiting the NF-κB/p38 kinase pathway in a dose-dependent manner (Ryu et al., 2010a,b). The chemical composition of seahorses will be affected by the extraction method, and most research is based on the freeze-dried powder and alcohol extract of seahorses (Mundijo et al., 2022; Ryu et al., 2010b), with limited studies conducted on seahorses decoctions.

Danio rerio (zebrafish) is a representative model for translational studies on human bone diseases. Particularly, the zebrafish embryo is a powerful model to study developmental osteogenesis (Kimmel et al., 2010). Zebrafish can absorb drugs or chemicals through the skin and are transparent in the embryo and young fish, which makes the internal organs of zebrafish directly observable to study the morphological changes (Barrett et al., 2006; Goldsmith, 2004; Kari et al., 2007; Zon and Peterson, 2005).

The ossification mechanism and bone matrix composition of zebrafish are highly similar to those of mammals, and zebrafish genomes also contain homologous genes to several key genes adjusting skeletal development in mammals (Dietrich et al., 2021). Thus, zebrafish can be used to make models of osteoporosis, osteogenesis imperfect, and fibrodysplasia ossificans progressive (de Vrieze et al., 2014; Fisher et al., 2003; LaBonty et al., 2017). Prednisolone can induce a dose-dependent loss of bone matrix (glucocorticoid-induced osteoporosis, GIOP) that can be assessed in zebrafish scales through histological and biochemical examinations (de Vrieze et al., 2014). Therefore, we used prednisolone to construct a GIOP model in zebrafish larvae, and the purpose of this study was to test the ability of seahorses decoctions to prevent the formation of the osteoporotic phenotype in the zebrafish GIOP model.

2. Materials and methods

2.1. Ethics statement

This experimentation has been done in the Zebrafish Laboratory (Sun Yat-sen University, guangzhou, China). All experimental protocols were approved by the Institutional Animal Care and Ethics Committee of Sun Yat-sen University.

2.2. The preparation of H. erectus decoction and component detection

Hippocampus erectus (*H. erectus*) were derived from HAIMAO SEED TECHNOLOGY GROUP CO., TLD, Zhanjiang, which were 6 months old (body length 9–11 cm and wet weight 7–9 g). The seahorses were anesthetized with tricaine methanesulfonate (MS-222, 100 mg/L), and then frozen with liquid nitrogen. Seahorses decoction (DHe) was made by 50 g seahorses and 5 L pure water. The preparation of seahorses decoction was referred to the preparation of Chinese herbal medicine decoction (Song et al., 2004). The liquid in the decoction was filtered through qualitative filter paper, and the volume was fixed to 5 L with pure water in a volumetric flask to prepare 10 g/L DHe.

Qualification of chemical markers of DHe were performed on a Q Exactive Plus system (Thermo Fisher, America). The mixtures were filtered by a 0.22 μm filter before separated by an Acquity UPLC HSS T3 column (1.8 μm, 2.1 mm × 100 mm). The mobile phase composed of 0.1 % formic acid in acetonitrile (A) and 0.1 % formic acid in water (B), and the gradient program according to the Kenneth et al. method (Kwan et al., 2019). The column temperature was 35 °C, and the injection volume was 10 μl. Mass spectrometry was performed on a Q Exactive Orbitrap equipped with an ESI ion source was operated in positive and negative ion. The dry gas temperature was 320 °C, drying gas flow rate was 15 L/min, and capillary voltage was 3200 V.

2.3. Animals husbandry and experimental design

Zebrafish larvae (*D. rerio*) AB strains were divided into six groups, each group was soaked with different solutions for 120 h in 4 days after fertilization (dpf) and set three parallels ($n = 30$). Control groups, larvae were treated with 0.1 % dimethyl sulfoxide (DMSO). Model groups, larvae were treated with 25 μM prednisone (Huo et al., 2018). Positive groups, larvae were treated with 25 μM prednisone and 30 mg/L etidronate disodium (ED). Low concentration seahorses decoction group (L-DHe), larvae were treated with 25 μM prednisone and 0.15 g/L DHe. Medium concentration seahorses decoction group (L-DHe), larvae were treated with 25 μM prednisone and 0.30 g/L DHe. High concentration seahorses decoction group (L-DHe), larvae were treated with 25 μM prednisone and 0.60 g/L DHe. All groups larvae were raised in closed water system (1 L) under 14/10 h light/dark (L/D) cycles at 28°C and fed with paramecia, and treatment solutions were changed every 24 h. Larvae were euthanized for sampling in 0.1 % 2-phenoxyethanol.

2.4. Skeletal staining, mineralized area and mineralized intensity in zebrafish larvae

Zebrafish larvae were collected and fixed in 2 % paraformaldehyde (PFA) overnight and trypsinized with 1 % trypsin in a 35 % saturated sodium borate solution overnight to remove the skin and muscles. After washing with 10 % glycerol/0.5 % KOH, larvae were stained with 0.02 % Alizarin Red stain/10 % glycerol/0.5 % KOH overnight, and then washed twice with 50 % glycerol/0.1 % KOH overnight. Three zebrafish larvae were selected from each group and three parallels were set. Images were acquired with a stereomicroscope (M165FC; Leica, Frankfurt, Germany). Digital image analysis was performed to quantify stained area and staining density.

Whole-mount skeletal staining of zebrafish larvae was referred to Ding et al. (2021). Images were acquired with a stereomicroscope (ZEISS Lumar.V12, Germany) under the bright field C4 brightness, and the exposure was maintained at 280 ms to obtain the abdominal surface view photos of juvenile zebrafish. Images were analyzed with Image Pro Plus 6.0 to calculate the mineralized area and mineralized intensity (Barrett et al., 2006).

2.5. RNA isolation, cDNA synthesis and quantitative real-time PCR

Total RNA was extracted from a pool of 10 larvae using Trizol (Invitrogen) reagent and reverse transcribed into cDNA with Superscript III Reverse Transcriptase (Invitrogen). Quantitative Real-Time PCR (qRT-PCR) was performed using an Roche LightCycler 480 instrument with the SYBR (TaKaRa, China) system with 40 cycles of 95 °C for 10 s, and 58 °C for 30 s. Each qRT-PCR assay was conducted with three independent biological samples and each sample was performed in triplicate. All expression levels were normalized to the expression level of the housekeeping gene β-actin. Relative mRNA expression levels are expressed as 2ΔΔCt. Primer sequences are listed in the supplementary information 1.

2.6. ALP and TRAP assays

Each group was set up with three replicates treatment, and 5 fish were used for each treatment.

Alkaline phosphatase (ALP) assay was done using an alkaline phosphatase assay kit (Njjcbio, China). Tartrate resistant acid phosphatase (TRAP) assay was performed using a tartrate-resistant acid phosphatase stain kit (Njjcbio, China) following the manufacturer's protocol.

2.7. Statistics

Data are presented as mean±s.d. Statistical analyses were performed with either an analysis of variance (ANOVA) or an unpaired two tailed Student's *t*-test. All statistical analyses were performed using SPSS 21.0 software (IBM) and values of P <0.05 were considered statistically significant.

3. Results

3.1. The main active components of DHe

There were 36 active components in DHe, and the main ingredients (relative content >2 %) were shown in Table 1. The 36 active ingredients contained 6 kinds of dipeptides (30.915 %), 5 kinds of amino acids (23.709 %), 10 kinds of organic acids (17.639 %), 5 kinds of organic acid esters (4.567 %), 2 kinds of vitamins (2.535 %), alcohol (2.162 %), 2 kinds of aldehydes (1.296 %), 2 kinds of nucleosides (0.439 %), ketone (0.240 %) and 2 kinds of other substances (creatinine and oleamide, total relative content of 16.497 %) (supplementary information 2).

3.2. Effect of DHe concentration on survival rate of zebrafish

To explore the appropriate concentration of DHe, 4dpf larva zebrafish were exposed to different concentrations of DHe. The survival numbers of larvae after 24 h, 48 h, 72 h, 96 h, 120 h were counted (Table 2) and the median lethal concentration of DHe was obtained (Table 2). The results showed that 0.01–0.3 g/L DHe had no obvious lethal effect on larvae (Table 2). However, when the DHe concentration exceeded 1 g/L, the mortality rate of larvae was positively correlated with concentration. With the increase of time, the survival slope of larva zebrafish decreased (Fig. 1 A) and the median lethal concentration of DHe on larvae decreased (Table 2).

According to the above results, we further explored the effects of DHe concentrations on the mortality of larvae under the GIOP model. On the fourth day, the mortality of larvae in all groups increased sharply except the control groups (Fig. 1B). During the whole experiment, the mortality rate of L-DHe, M-DHe and H-DHe groups were 20.8 %, 17.5 % and 21.7 % respectively (Table 3), which was close to that of the model groups (22.5 %). It was showed the death of larvae was mainly caused by prednisolone, and 0.15 g/L, 0.3 g/L, 0.6 g/L DHe had no obvious lethal effect on osteoporotic larvae.

3.3. DHe treatment stimulates osteogenesis in zebrafish

Staining with alizarin red was performed to evaluate the mineralization area and degree of bone mineralization, including spine, palatoquadratecartilage (PQ), occipital bone (OC), parasphenoid (PS), branchiostegal (BS), operculum (OP), notochord (NC), ceratohyal (CH), cleithrum (CL) and otolith (OT). The results showed that prednisolone inhibited the degree and area of mineralization in all bones except OT (Fig. 2 A)

The results showed that prednisolone could effectively inhibit bone mineralization and reduce the mineralized area of larvae. For the spine, no mineralization was observed in model groups, while the mineralization was observed in positive groups, L-DHE and M-DHe groups (Fig. 2B and C), indicating that 0.15 g/L and 0.30 g/L DHe could improve the inhibition effect of prednisolone on the spine.

Table 1
The main active components of DHe.

Name	Molecular formula	Molecular formula (%)
Glycylproline	$C_7H_{12}N_2O_3$	24.562
Creatinine	$C_4H_7N_3O$	12.852
L-Phenylalanine	$C_9H_{11}NO_2$	12.296
Cholic Acid	$C_{24}H_{40}O_5$	6.725
L-Leucine	$C_6H_{13}NO_2$	6.288
Succinic Acid	$C_4H_6O_4$	5.940
Oleamide	$C_{18}H_{38}NO$	3.645
Leucylproline	$C_{11}H_{20}N_2O_3$	3.029
Taurine	$C_2H_7NO_3S$	2.927
Bis Sorbitol	$C_{24}H_{30}O_6$	2.162
Dibutyl Phthalate	$C_{16}H_{22}O_4$	2.127

Table 2
Survival number of larva zebrafish (n =90) and median lethal concentration of DHe.

Time (h)	DHe concentration (g/L)								LC50 (g/L)
	0	0.01	0.03	0.1	0.3	1	3	10	
24	90	90	90	90	90	90	44	0	2.990
48	89	90	90	90	90	87	42	0	2.881
72	89	90	90	90	90	86	41	0	2.828
96	89	90	90	89	90	85	38	0	2.694
120	89	90	89	88	90	84	35	0	2.559

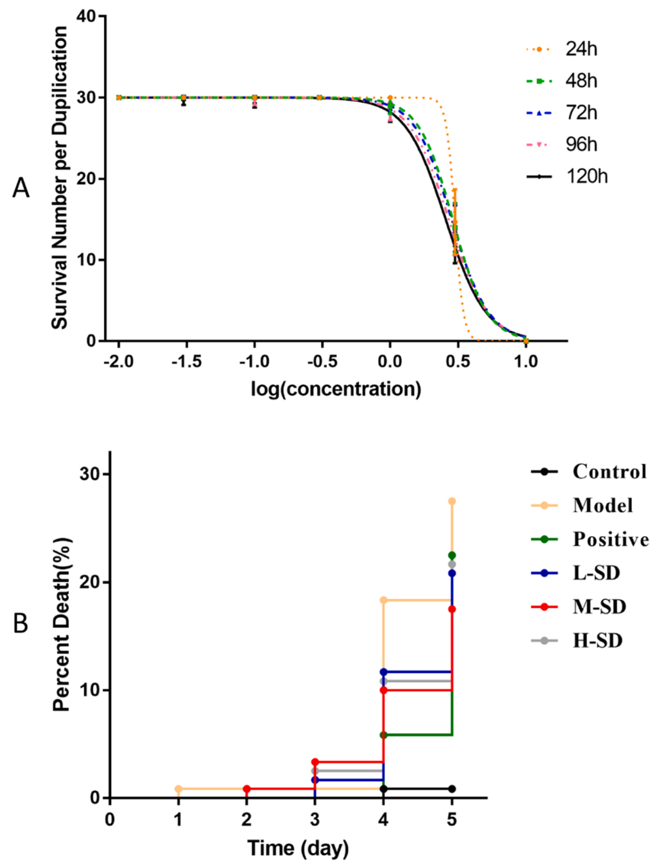

Fig. 1. (A) The effect of seahorse decoction on survival rate of zebrafish larvae. (B) The effect of seahorse decoction on mortality of osteoporosis zebrafish larvae. L-DHe, low concentration (0.15 g/L) seahorse decoction groups. M-DHe: medium concentration (0.3 g/L) seahorse decoction groups. H-DHe: high concentration (0.6 g/L) seahorse decoction groups.

Table 3
The total mortality of larva zebrafish in each group (n=90).

Groups	Control	Model	Positive	L-DHe	M-DHe	H-DHe
Mortality	0.8 %	27.5 %	22.5 %	20.8 %	17.5 %	21.7 %

Similarly, the mineralization degree and area of PS, BS, and OP in M-DHe groups were significantly higher than those in model groups ($P < 0.05$) (Fig. 2B and C), indicating that 0.15 g/L DHe could weaken the inhibitory effect of prednisolone.

The results of each group were quantitatively analyzed, and the total staining area and staining integral optical density of larvae in each group were calculated. ED could significantly increase the calcified area ($P < 0.001$) and the degree of calcification ($P < 0.05$) (Fig. 2D). These results indicated that the osteoporosis induced by prednisolone can be effectively improved by DHe. 0.3 g/L DHe can significantly increase the

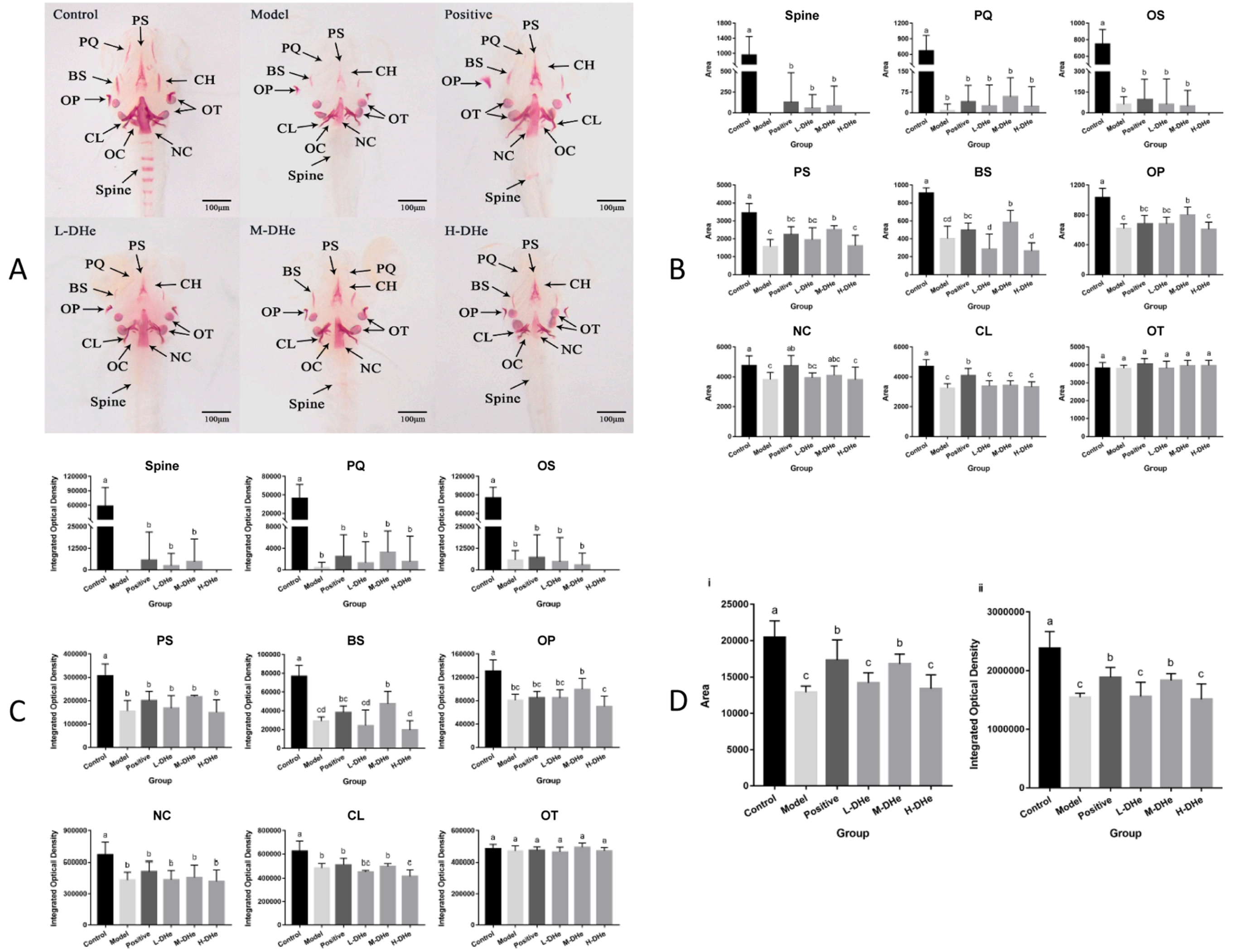

Fig. 2. (A) The influence of seahorse decoction on alizarin red staining in osteoporosis zebrafish larvae. L-DHe, low concentration (0.15 g/L) seahorse decoction groups. M-DHe: medium concentration (0.3 g/L) seahorse decoction groups. H-DHe: high concentration (0.6 g/L) seahorse decoction groups. Palatoquadratecartilage (pq), occipital bone (oc), parasphenoid (ps), branchiostegal (bs), operculum (op), notochord (nc), ceratohyal (ch), cleithrum (cl) and otolith (ot). The results were taken at 50× stereoscopic microscope in bright field. (B) The influence of seahorse decoction on bone mineralization area in osteoporosis zebrafish larvae of each bone. Different lowercase indicates significant difference between each group ($P < 0.05$). (C) The influence of seahorse decoction on bone mineralization level in osteoporosis zebrafish larvae of each bone. Different lowercase indicates significant difference between each group ($P < 0.05$). (D) The influence on bone mineralization in osteoporosis zebrafish larvae of seahorse decoction. Different lowercase indicates significant difference between each group ($P < 0.05$).

degree of mineralization of larvae ($P < 0.01$), and the effect is similar to that of ED.

3.4. Expression of ALP, TRAP and bone development genes

Comparing with the model groups, the positive groups significantly increased the activity of ALP ($P < 0.01$) and decreased the activity of TRAP ($P < 0.0001$) (Fig. 3 A), indicating that after treatment with positive drugs, osteoblast activity increased and osteoclast activity decreased, thus successfully ameliorating osteoporosis. 0.3 g/L and 0.6 g/L DHe significantly increased the expression activity of ALP ($P < 0.01$), 0.15 g/L DHe could increase the expression activity of ALP, but there was no significant difference compared with model groups ($P > 0.05$). 0.15 g/L and 0.3 g/L DHe can reduce the TRAP activity of larvae ($P < 0.0001$), but 0.6 g/L DHe had little inhibitory effect on TRAP (Fig. 3 A). In conclusion, the effect of 0.3 g/L DHe on larvae osteoporosis was consistent with that of ED, indicating that the concentration of DHe had an improving effect on osteoporosis.

The expressions of *alp*, *acp5a*, *o*cn and *rankl* in larvae were significantly reduced in model groups (Fig. 3B), indicating that 25 μM prednisolone inhibited the expression of these genes. The ED promoted the expression of *alp* and *o*cn, but the expression levels of these two genes could not recover to control groups. The expression of *alp* and *o*cn in M-DHe groups was higher than that in L-DHe and H-DHe groups, (Fig. 3B) showing that 0.3 g/L DHe had a greater promoting effect on osteogenesis than 0.15 g/L and 0.6 g/L DHe.

4. Discussion

Seahorses are rich in amino acids, fatty acids and trace elements. The phenolic compounds extracted from hippocampus can promote the development of osteoblasts (Yuan et al., 2018). The alcohol extracts can remove hydroxyl radical, superoxide radical and alkyl radical, inhibit intracellular reactive oxygen species (Qian et al., 2008). The enzymic extract can increase the time of loading exercise, reduce the production of lactic acid and urea nitrogen during exercise, promote the elimination of lactic acid and urea nitrogen after exercise (Guo et al., 2017). Studies have shown that different dosage forms of Chinese medicine contain

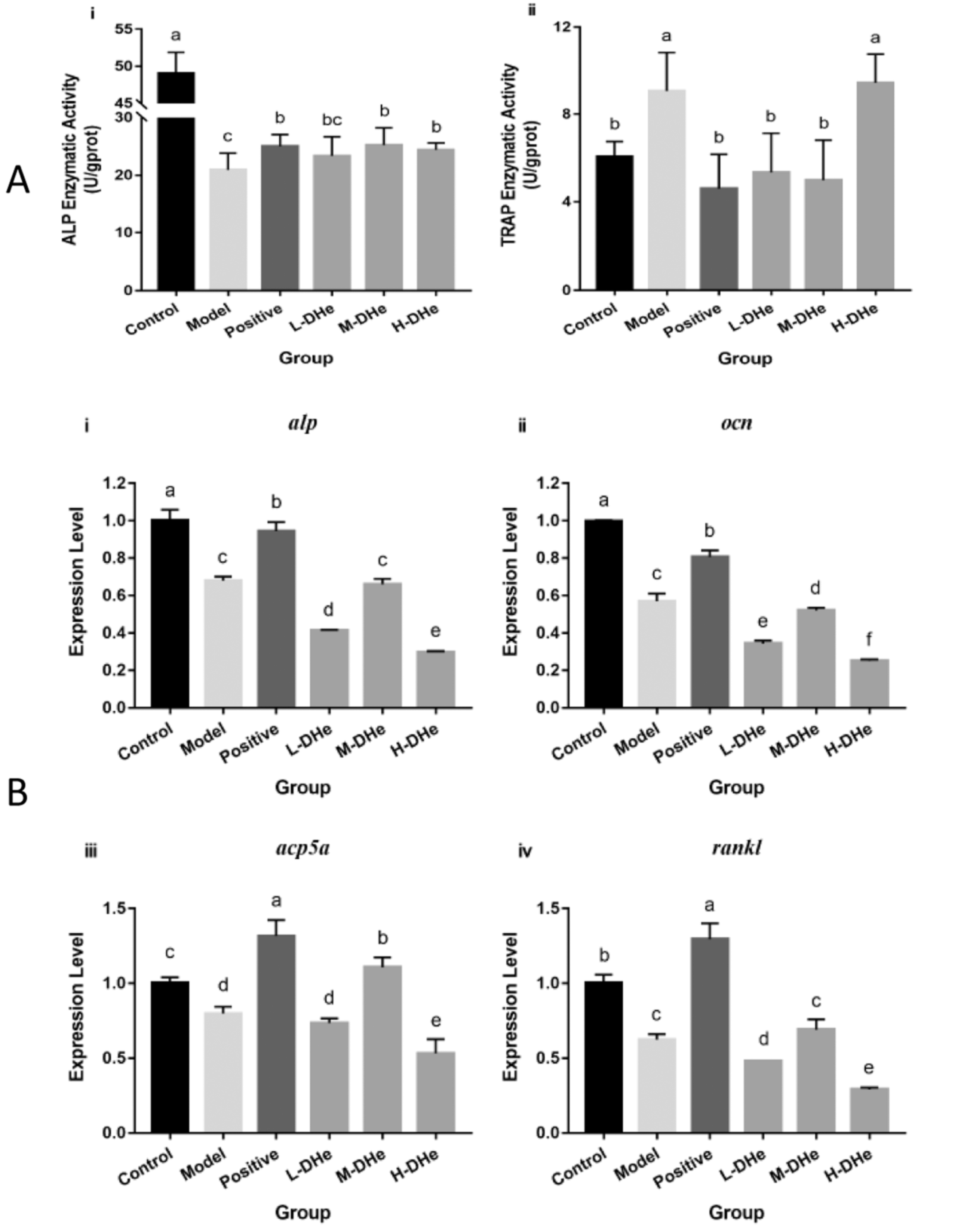

Fig. 3. **(A)** The influence of seahorse decoction on the expression of bone-related enzyme activity in osteoporosis zebrafish larvae. L-DHe, low concentration (0.15 g/L) seahorse decoction groups. M-DHe: medium concentration (0.3 g/L) seahorse decoction groups. H-DHe: high concentration (0.6 g/L) seahorse decoction groups. i: the expression of ALP; ii: the expression of TRAP. Different lowercase indicates significant difference between each group ($P < 0.05$). (B) The influence of seahorse decoction on the expression of genes related to bone development in osteoporosis zebrafish larvae. i: the expression of *alp*; ii: the expression of *acp5a*; iii: the expression of *o*cn ; iv: the expression of *rankl*. Different lowercase indicates significant difference between each group ($P < 0.05$).

different active ingredients and content (Chen et al., 2009; Zhang et al., 2019). There were 6 kinds of dipeptides in DHe, the total relative content was 30.915 %, among which glycylproline accounted for the highest proportion among all active ingredients, the relative content reached 24.562 %. The relative content of oleamide and ethyl docosahexaenoate was 3.645 % and 0.689 %, The relative content of was. which were different from the composition of seahorse freeze-dried powder (Li et al., 2022). There were many components in DHe that were related to bone development, such as taurine and succinic acid (Table 1). Taurine increased the ALP activity and mineralized nodules in human mesenchymal stem cells induced by osteogenic induced medium, and up-regulated the expression of ALP, osterix and osteopontin in a dose-dependent manner (Zhou et al., 2014).

Succinic acid may inhibit the proliferation and osteogenic differentiation of bone precursor cells in mice by inhibiting the expression of *Runx2* and *O*CN (Wei et al., 2019).

Zebrafish is an ideal model which possesses many advantages. The bone elements, ossification mechanism and bone matrix components of zebrafish share many common characteristics with mammals, which makes zebrafish model widely used in the study of bone diseases (Barbazuk et al., 2000). Importantly, 82 percent of human disease-related genes can be found in zebrafish, including genes associated with bone development (Li et al., 2009). It has been reported that there are few TRAP-positive osteoclasts observed in zebrafish larvae until 20dpf, and there is few bone remodeling in zebrafish larvae (Chen et al., 2014). Therefore, zebrafish larvae might be more suitable for the research and development of anabolic agents for GIOP than other models (Luo et al., 2016).

High concentration of Chinese herbal medicine has certain toxicity (Carnovali et al., 2021). We found the dibutyl phthalate (DBP) in DHe (Table 1), which sharply increased the mortality rate of zebrafish when DBP concentration was higher than 0.05 mg/L (Pu et al., 2020). Different from the traditional medication mode, because zebrafish can absorb the drug through the skin (Zon and Peterson, 2005), our experiment adopts the method of immersion. The results showed that the medial lethal concentration (LC_{50}) of DHe on zebrafish was not directly proportional. The lethal effect of 0.01–0.3 g/L DHe on zebrafish was not significant. But when the concentration of DHe exceeded 1 mg/L, the survival rate of zebrafish would decline rapidly (Table 2).

ED, as a medicine for osteoporosis, can promote the proliferation and differentiation of osteoblasts (Barrett et al., 2006). *acp5a* encodes TRAP which is commonly used as an osteoclast marker (Khrystoforova et al., 2022), and *rankl* has been confirmed to be the main regulator of osteoporosis (Zhao et al., 2020). The results showed that ED could effectively alleviate the osteoporosis induced by prednisolone and promote the expression of *alp*, *o*cn, *acp5a* and *rankl*. 0.3 g/L DHe significantly increased the degree of bone mineralization in zebrafish larvae, increased the expressions of ALP and *acp5a*, and inhibited the expression of TRAP (Fig. 1D), indicating that prednisolone-induced osteoporosis could be inhibited. The degree and area of M-DHE mineralization were the same as those in the positive groups, and were significantly higher than those in the other groups, indicating that the effects of 0.3 g/L DHe and ED were similar. However, 0.15 g/L and 0.6 g/L DHe had no significant improvement effect on osteoporotic zebrafish larvae. There was no significant difference expression of TRAP between the H-DHe groups and the model groups, indicating that 0.6 g/L DHe had no inhibitory effect on TRAP.

Chinese medicine contains many components, and each component is composed of multiple components. Traditional Chinese medicine has a wide range of targets and cannot be evaluated by the content of one or some of its components. In this study, it was found that there are many active ingredients in DHe, and some of these active ingredients had opposite effects. Taurine has a protective effect on bone loss and promotes collagen synthesis in rats (Park et al., 2001), but succinic acid and dibutyl phthalate may cause low bone density and affect bone development (Pu et al., 2020). The effect of some components decreases with the increase of concentration, for example, taurine can inhibit the differentiation of aortic valve stromal cells into osteoblasts at a dose of 10 μM, and 20 μM is less effective than 10 μM (Zhou et al., 2014). Similarly, there are some ingredients that do not have a single effect, oleamide can both inhibit the development of osteoblast and osteoclast (Ciovacco et al., 2009; Ransjö et al., 2003; Schiller et al., 2001). All of these components were found in DHe. Interestingly, the expression of *alp*, *o*cn, *acp5a* and *rankl* in the H-DHe groups were lower than that in the L-DHe, and the expression of these genes in both groups was significantly lower than that in the model groups. It is speculated that antagonistic substances inhibit the expression of these genes with the increase of DHe concentration.

5. Conclusions

In this study, oleamide, docosahexaenoic acid and other new components were detected, which enriched the active components of seahorses. Meanwhile, it was found that 0.3 g/L DHe significantly increased the degree of bone mineralization, promoted the activity of ALP and inhibited the activity of TRAP, promoted the expression of osteogenic genes, and improved the osteoporosis of zebrafish larvae, but its molecular mechanism needs to be further studied. Overall, this study provides a theoretical basis for DHe treatment of osteoporosis.

CRediT authorship contribution statement

Jinhui Wu: Writing – original draft, Visualization, Validation. **Li Lin:** Formal analysis, Data curation, Conceptualization. **Guifeng Li:** Investigation, Funding acquisition, Formal analysis. **Chenlei Huang:** Resources, Project administration, Funding acquisition. **Ting Wang:** Methodology. **Jiahui Chen:** Project administration, Investigation, Funding acquisition. **Zhilun Zhang:** Funding acquisition, Data curation, Conceptualization. **Yong Zhang:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Xuange Liu:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Qiuxian Chen:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Huadong Yi:** Methodology.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This research was supported by the Southern Marine Science and Engineering Guangdong Laboratory (Zhuhai) (SML2023SP234), and Research on Breeding Technology of Candidate Species for Guangdong Modern Marine Ranching (2024-MRB-00–001).

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.aqrep.2024.102510.

Data availability

Data will be made available on request.

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